RAHUL KP

Applied Machine Learning Engineer | AI/ML Engineer

Kerala, India · Open to Relocation rahulkpkurup@gmail.com · GitHub · LinkedIn · Portfolio · Hugging Face · Google Scholar · ORCID

SUMMARY

Applied Machine Learning Engineer with an **MSc in Computer Science** and hands-on experience building machine learning systems from mathematical foundations through deployment. Strong background in **Python, PyTorch, deep learning, numerical computing, ML pipelines, and AI system development**, with projects spanning neural networks, computer vision, recommendation systems, optimization-oriented modeling, and production ML APIs. Interested in applying machine learning to **engineering and scientific design automation problems**, including data-driven optimization and autonomous AI workflows.

EDUCATION

MSc Computer Science

University of Calicut · 2024–2026 **77%**

BSc Computer Science

University of Calicut · 2020–2023

TECHNICAL SKILLS

Programming: Python, C++, C, Java, SQL, JavaScript/TypeScript,

Machine Learning: Supervised Learning, Regression, Classification, Feature Engineering, Model Evaluation, Hyperparameter Optimization

Deep Learning: Neural Networks, CNNs, Backpropagation, Automatic Differentiation, PyTorch, NumPy

AI Systems: LLMs, RAG, LangChain, FastAPI, Prompt Engineering

Optimization & Mathematics: Linear Algebra, Calculus, Matrix Algebra, Numerical Methods, Gradient-Based Optimization

ML Engineering: Docker, Kubernetes, Git, GitHub Actions, REST APIs, PostgreSQL, Redis

---

SELECTED PROJECTS

ANN-Foundation — Neural Network & Autograd Engine

Python · NumPy · Automatic Differentiation · Deep Learning

- Built a neural-network framework **from scratch in Python without deep-learning frameworks**, implementing reverse-mode automatic differentiation and computational graphs.
- Implemented core components including **tensor operations, gradients, layers, losses, optimizers, and MLP training**.
- Extended the framework toward computer vision with convolutional and pooling operations, demonstrating understanding of the mathematical foundations behind modern deep learning systems.
- Verified analytical gradients against **numerical finite-difference approximations**, validating the correctness of the implementation.
- Repository: github.com/rahulkp-ai/ann-foundation

NCF-Recommender-System

Python · NumPy · PyTorch · Neural Collaborative Filtering

- Implemented a **Neural Collaborative Filtering** recommendation system using both a NumPy-based implementation and PyTorch benchmark.
- Developed an end-to-end recommendation pipeline covering **data processing, model construction, training, evaluation, and inference**.
- Studied the relationship between embedding representations, neural architectures, and recommendation performance through reproducible experiments.
- Research preprint: DOI · Repository: github.com/rahulkp-ai/NCF-Recommender-System

PhishGuard — Production ML System

Python · Flask · Random Forest · Docker · Kubernetes · CI/CD

- Built a production-oriented phishing URL detection system using **28 engineered URL features** and machine-learning classification.
- Developed the complete pipeline from feature extraction and model training to **REST API deployment**.
- Containerized the application with Docker and configured Kubernetes deployment with automated CI/CD.
- Achieved **98%+ test coverage**, emphasizing reliability and maintainability of the ML application.
- Repository: github.com/rahulkp-ai/phishguard

RESEARCH

Neural Collaborative Filtering for Recommendation Systems

Research Project / Preprint

- Investigated neural recommendation architectures and evaluated a reproducible implementation of Neural Collaborative Filtering.
- Explored model architecture, representation learning, training methodology, and empirical evaluation.
- Published as a research preprint with DOI: 10.20944/preprints202605.0449.v1

CERTIFICATIONS

- IBM Generative AI Engineering Professional Certificate - IBM
- MLOps — Machine Learning Operations Specialization - Duke University
- Neural Networks and Deep Learning
- Transformer Models and BERT — Google Cloud
- Advanced C++ & Modern Practices — Microsoft
- Matrix Algebra for Engineers — HKUST